

$$\Delta R = (\max R - \min R)/2 = (163.4 - 156.6)/2 = 3.4 \text{ s}$$

2 **Ans: C**

time to accelerate to $20 \text{ m s}^{-2} = 20 / 0.2 = 100 \text{ s}$

time to decelerate to rest = $20 / 0.4 = 50 \text{ s}$

distance travelled = $\frac{1}{2} (20)(100) + 20(t) + \frac{1}{2} (20)(50)$

$3000 = \frac{1}{2} (20)(100) + 20(t) + \frac{1}{2} (20)(50)$

$t = 75$

total time taken = $100 + 75 + 50 = 225 \text{ s}$

2

3 **Ans: D**